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Testování efektu AI Foundational models na analýzu NGS dat DNA sekvenování

Diplomová práce

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Abstrakt

V této bakalářské/diplomové/rigorózní práci se věnujeme …

Abstract

In this thesis we study ...

Místo tohoto listu vložte kopii oficiálního zadání práce bez podpisů.

Poděkování

Na tomto místě bych chtěl(-a) poděkovat …

Prohlášení

Prohlašuji, že jsem svouji diplomovou práci vypracoval samostatně pod vedením vedoucího práce s využitím informačních zdrojů, které jsou v práci citovány. Prohlašuji, že jsem pro stylistické úpravy textu využil nástroje AI a to plně v souladu s principy akademické integrity.

Brno 13. Května 2026

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Tomáš Vamberský

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List of abbreviations

$\mathbb{C}$ množina všech komplexních čísel
$\mathbb{R}$ množina všech reálných čísel
$\mathbb{Z}$ množina všech celých čísel
$\mathbb{N}$ množina všech přirozených čísel

Introduction

Chapter 1

Theoretical background

1.1 Transcription factor binding

1.1.1 Transcription factors and DNA binding

Transcription factors (TFs) are proteins that modulate transcription by binding DNA in a sequence-specific manner. They act through regulatory elements such as promoters, enhancers, and silencers, recruiting the transcription machinery or modulating its activity. Most TFs share a modular architecture: a DNA-binding domain (DBD) that contacts the target sequence, paired with one or more effector domains responsible for transactivation, repression, or dimerisation. Because the DBD determines binding specificity, TFs are classified by DBD family rather than by effector function. The human genome encodes approximately 1600 sequence-specific TFs distributed across roughly 75 DBD families (Lambert et al., 2018). The families represented among the TFs studied in this thesis include the C2H2 zinc fingers (the largest human family, exemplified by CTCF), the basic helix-loop-helix proteins (MYC, which binds as an obligate heterodimer with MAX), and the ETS domain family (SPI1).

Target sequences are typically short motifs of 6–20 bp, recognized with tolerance for sequence variation. Despite the considerable structural variety across DBD families, all sequence-specific TFs face the same fundamental problem: they must locate their short target sequence among a large number of similar but non-functional sites distributed across a genome of several gigabases.

1.1.2 Sequence determinants of binding

The specificity of protein–DNA recognition rests on two complementary mechanisms: direct chemical recognition of bases (base readout) and recognition of sequence-dependent DNA geometry (shape readout).

Base readout operates through direct contacts — hydrogen bonds and van der Waals interactions — between amino acid side chains and the Watson–Crick edges of bases, predominantly in the major groove. *The major groove presents a distinctive pattern of hydrogen bond donors and acceptors* that permits discrimination between all four base pairs, including the orientation of A:T and T:A (Luscombe et al., 2001; Seeman et al., 1976). The C2H2 zinc finger illustrates the mechanism: each $\beta\beta\alpha$ module contacts approximately three base pairs through specificity residues on the recognition helix, and the contributions of successive modules combine to determine the affinity and specificity of the full array (Pavletich a Pabo, 1991). Position weight matrices (PWMs) are the natural statistical encoding of base readout, treating each position within the motif as an independent contribution to binding affinity (§1.3).

Shape readout depends on sequence context beyond the contact position. The geometry at any given base pair — minor groove width, propeller twist, helical twist, roll, and slide — is influenced by neighbouring bases over several positions in either direction, and is read by protein contacts that respond to local shape and electrostatics rather than to specific bases. The canonical example is recognition of narrow minor grooves by arginine residues: runs of A:T base pairs (A-tracts) produce locally narrowed minor grooves with concentrated negative electrostatic potential, which arginine side chains insert into and stabilise without forming base-specific hydrogen bonds (Rohs et al., 2009).

The two mechanisms are genuinely complementary rather than redundant. Crystal structures of TF–DNA complexes routinely show major-groove base-specific contacts and minor-groove shape-based contacts within the same interface, and augmenting PWM-based models with predicted shape features systematically improves binding prediction across TF families (Mathelier et al., 2016). Their combined reach is nonetheless bounded by the binding footprint: base readout depends on direct contacts within the 6–20 bp footprint, and shape readout, while sensitive to flanking sequence, is mediated through contacts within and immediately adjacent to it (Yang et al., 2017).

1.1.3 The specificity paradox

The degeneracy of TF motifs means that a typical 10 bp motif at moderate information content matches millions of sites in a mammalian genome, while in vivo occupancy — assayed by ChIP-seq, the standard method described in §1.2 — typically identifies between a few thousand and a few tens of thousands of bound sites per TF per cell line. **[TODO: cite ENCODE probably]** CTCF in K562 cells, for example, has approximately 52 000 IDR-thresholded ENCODE peaks, against approximately 870 000 matches to the canonical CTCF motif in the hg19 genome at standard FIMO thresholds (Dozmorov et al., 2022). This discrepancy between motif presence and actual occupancy has been termed the transcription factor specificity paradox; **[TODO: if keep, reformulate, cite (Kribelbauer et al., 2019)]**In practice, the paradox is reflected in the weak correlation between intrinsic in vitro affinity (measured by HT-SELEX or protein-binding microarrays) and in vivo occupancy: high-affinity sites can remain unbound, while low-affinity sites are often occupied (Slattery et al., 2014).

Several mechanisms partially resolve the discrepancy, each a substantial field in its own right. Chromatin accessibility is regarded as the dominant factor: only a small fraction of the genome is accessible to DNA-binding proteins at any given time, the remainder occluded by nucleosomes or compacted into heterochromatin, and most in vivo TF binding occurs within accessible regions (Thurman et al., 2012). Cooperative binding with cofactors restricts binding further: many TFs bind productively only as part of a complex — homodimers, heterodimers, or larger assemblies — so the combinatorics of cofactor co-occupancy limit where productive binding can occur (Sönmezer et al., 2021). Cellular context adds a third layer: TF abundance, post-translational modification, and subcellular localisation are all condition-dependent and regulate the effective concentration of binding-competent protein (Spitz a Furlong, 2012). **[TODO: find alternative reference ideally]**

A fourth contribution, less studied than the preceding three, is sequence context beyond the motif itself. Dror et al. (2015) demonstrated, in a systematic analysis of 239 TFs from HT-SELEX and 56 from ChIP-seq data, that the sequence and DNA shape environment surrounding bound motifs differs significantly from that surrounding unbound occurrences of the same motif, with differences extending well beyond the core binding site and exhibiting TF-family-specific patterns. **[TODO: distinguish from shape readout somehow, read dror again]** Crucially, the signal was present in both the in vitro and in vivo datasets, ruling out the cellular environment as its sole source and establishing flanking sequence context as a genuine, sequence-encoded contributor to binding discrimination. Unlike chromatin accessibility and cooperative binding, this contribution is in principle directly accessible to any model that operates on raw sequence input.

The spatial scale at which sequence context influences binding spans several orders of magnitude: from the immediate flanks of the core motif (Dror et al., 2015), through the tens-to-hundreds of base pairs over which sequence composition modulates the nonspecific affinity experienced by a scanning TF (Afek a Lukatsky, 2013; Sela a Lukatsky, 2011), *out to the kilobase-scale compositional gradients recently characterised by Faltejsková et al. (2026)*. That last study, and its role as the biological motivation for the analyses in this thesis, is discussed in §1.2.8 following the description of the experimental data on which both studies are based. **[TODO: soften claim - descriptive? implied?, is this first mention? mention it's a preprint?]**

1.2 Next-generation sequencing and ChIP-seq

1.2.1 Short-read sequencing

Next-generation sequencing (NGS) refers to high-throughput DNA sequencing technologies that determine the nucleotide sequence of millions to billions of short DNA fragments in parallel. The dominant platform in current use is Illumina sequencing-by-synthesis: libraries of DNA fragments are clonally amplified on a flow cell surface, and each cycle of synthesis incorporates fluorescently labelled reversible-terminator nucleotides whose identity is read by imaging (Bentley et al., 2008). Typical read lengths are 50–150 bp per end; paired-end protocols sequence both ends of each fragment and are standard for ChIP-seq. The output is a set of FASTQ files containing read sequences and per-base quality scores. Key limitations are the short read length, which complicates mapping to repetitive regions, and a characteristic per-base error rate of approximately 10^{-3}, dominated by substitution errors (Stoler a Nekrutenko, 2021).

NGS functions as a platform on which different upstream wet-lab protocols measure different genomic quantities: RNA-seq measures transcript abundance, ATAC-seq measures chromatin accessibility, and ChIP-seq measures protein–DNA contacts. The platform is otherwise identical across assays; what changes is the biological molecule subjected to fragmentation and *sequencing*.

1.2.2 Chromatin immunoprecipitation sequencing

Chromatin immunoprecipitation followed by sequencing (ChIP-seq) is the standard method for mapping genome-wide protein–DNA contacts at the population level (Park, 2009). **[TODO: no acess,]** In the crosslinking variant used for most TF ChIP-seq experiments, cells are treated with formaldehyde to covalently crosslink protein–DNA complexes in situ, chromatin is then fragmented by sonication or enzymatic digestion to a target size of approximately 200 bp, and an antibody specific to the TF of interest is used to immunoprecipitate the crosslinked complexes. After reversal of crosslinks, the co-purified DNA is sequenced. The resulting reads are enriched at genomic positions occupied by the protein, producing a coverage signal that is higher at bound sites than at unbound genomic background.

The output of ChIP-seq is interpretable as a population-averaged estimate of contact frequency: sites occupied in a large fraction of cells contribute more fragments to the library and produce sharper, higher-amplitude peaks. This population averaging is a fundamental limitation: the assay does not report single-cell occupancy, cannot distinguish direct from indirect binding (a TF pulled down as part of a complex is not directly bound to the immunoprecipitated DNA), and is sensitive to antibody quality in ways that are difficult to control post-hoc.

1.2.3 CUT&Tag as a methodological contrast

[TODO: cut or not? decide! not really relevant to thesis] An alternative to ChIP-seq that is increasingly used for TF mapping is CUT&Tag (Kaya-Okur et al., 2019). Rather than immunoprecipitating crosslinked complexes, CUT&Tag uses a Protein A–Tn5 transposase fusion, directed to the protein of interest by an antibody, to tagment (cut and insert sequencing adaptors into) the DNA in the immediate vicinity of the bound protein. The method requires lower cell inputs, produces lower background, and generates sharper peaks than ChIP-seq, making it particularly attractive for TFs with weak or transient binding. One CUT&Tag dataset (CTCF in K562 cells; (Kaya-Okur et al. 2019)) is included in the Faltejsková et al. dataset to which this thesis refers; all ChIP-seq data used in this thesis are from conventional ChIP-seq experiments.

1.2.4 Read alignment and peak calling

Sequencing reads are aligned to the reference genome using short-read aligners such as BWA (H. Li, 2013) or Bowtie 2 (**langmeadFastGappedreadAlignment2012**), **[TODO: verify how to cite bowtie - i dont have access]** producing coordinate-sorted BAM files with alignment information and mapping quality scores. Reads mapping to multiple locations are typically filtered; duplicate reads arising from PCR amplification of the same fragment are also marked and excluded.

Enriched genomic regions (peaks) are identified by peak-calling algorithms that compare local fragment density in the immunoprecipitated sample against a control library, usually input chromatin from the same cells. MACS2 is the standard algorithm for TF ChIP-seq (Zhang et al., 2008); it models the expected fragment distribution, estimates local background from a sliding window, and assigns a p-value and q-value to each candidate peak under a Poisson model. The primary output is a narrowPeak BED-format file containing one row per peak, with columns for chromosomal coordinates, a peak score, $-\log_{10}(p)$, $-\log_{10}(q)$, and the offset of the peak summit from the start coordinate. For TF ChIP-seq, peak widths are typically 200–500 bp.

1.2.5 Reproducibility filtering and IDR

Because ChIP-seq is sensitive to antibody efficiency and library complexity, peaks called from a single experiment may include irreproducible signals. The Irreproducible Discovery Rate (IDR) framework provides a principled way to rank peaks by their reproducibility across biological or technical replicates (Q. Li et al., 2011). IDR fits a mixture model to the joint rank distribution of peaks across two replicates and assigns each peak a probability of arising from the reproducible component of the mixture. Peaks below an IDR threshold are retained; those above are discarded as likely irreproducible. The ENCODE project distributes IDR-thresholded peak calls as the recommended file type for downstream analysis (Dunham et al., 2012); conservative IDR-thresholded calls, which are thresholded at an IDR of 0.05 applied to the self-consistency ratio as well as the pooled-consistency ratio, are preferred where available. All peak files used in this thesis are IDR-thresholded narrowPeak files obtained from the ENCODE portal **[TODO: maybe a cite]**.

1.2.6 Data formats and reference genome

The narrowPeak format used throughout this thesis is a ten-column extension of the BED format, with the extension columns carrying the MACS2 signal, p-value, q-value, and summit offset. **[TODO: a few example lines of a narrowPeak file with the columns, for illustration?]** Reference genome sequences are stored in FASTA format and indexed with `samtools faidx` to allow efficient random-access retrieval of arbitrary intervals, which is required for the window-extraction step described in §**??**.

This thesis uses the hg19/GRCh37 human reference assembly throughout, consistent with the Faltejsková et al. analysis. The hg38/GRCh38 assembly is the current reference standard and differs from hg19 primarily in gap-filling and a small number of chromosome-level corrections; coordinates are not directly interchangeable, and liftover is required to transfer annotations between assemblies. Non-B DNA and CpG island annotations used by Faltejsková et al. are hg38-based and were not used in this thesis.

1.2.7 The ENCODE project

The Encyclopedia of DNA Elements (ENCODE) project is a large-scale, consortium-driven effort to functionally annotate the human genome through systematic application of high-throughput assays including ChIP-seq, ATAC-seq, RNA-seq, and others

(Dunham et al., 2012). ENCODE enforces standardised experimental protocols, antibody validation requirements, and bioinformatic pipelines, and applies automated quality-control audits to all deposited experiments. Experiments flagged by these audits as failing quality thresholds are marked accordingly on the portal; no red-flagged experiments are included in this thesis. All data are publicly available through the ENCODE portal (`https://www.encodeproject.org`) under permissive data use agreements. **[TODO: decide if the url is a footnote]**

*The TFs and cell lines used in this thesis — MYC, CTCF, and SPI1 across cell lines GM12878, H1, K562, and MCF-7 — were selected from the Faltejsková et al. supplementary accession list, which is itself drawn from ENCODE. Full accession identifiers and selection criteria are given in §**??**.* **[TODO: consider dropping]**

1.2.8 Long-range sequence context around binding sites

The ChIP-seq peak calls distributed by ENCODE identify the genomic coordinates of protein–DNA contacts, but the narrowPeak format records only a 200–500 bp window around the occupied site. Whether the broader sequence context surrounding a bound site encodes systematic, non-random features — and whether those features are consistent with a role in guiding TF recognition — is a question that cannot be answered from the peak coordinates alone.

Faltejsková et al. (2026) addressed this question in a recent descriptive study published in *Nucleic Acids Research*. They analysed a ±5000 bp window around in vivo binding sites for ten TFs drawn from ENCODE ChIP-seq and one CUT&Tag dataset, dividing each 10 kb window into 50 bp non-overlapping segments and computing sequence composition features — GC content, dinucleotide frequencies, predicted DNA shape (via deepDNAshape), and HOCOMOCO-based TF affinity scores — across the profile from distal flanks to peak centre.

The principal finding is that GC content is significantly elevated in a patch spanning approximately 1–2 kb around the binding site for most of the ten TFs examined, with a typical magnitude of ~10 percentage points above the distal flank. Directional asymmetries in dinucleotide composition, pointing toward the binding site, were observed for several TFs including MYC, FOXK2, and p53, and are absent or weak for CTCF and SPI1. DNA shape features predicted from the local sequence context — in particular roll and helical twist — change in a manner consistent with increased bendability near the binding site. A small but significant enrichment of predicted Z-DNA and G-quadruplex motifs at the binding site was also observed for selected TFs, though the authors note these are insufficient to account for the full dinucleotide pattern.

These findings are interpreted through a "statistical funnel" model: broad compositional gradients around binding sites create a low-free-energy landscape that biases TFs undergoing one-dimensional facilitated diffusion toward the target site, complementing the short-range specific recognition described in §1.1.2. The study explicitly characterises this model as hypothesis-generating. Critically, it is a descriptive analysis: no model is tested for whether it can detect or exploit the sequence gradients described.

This descriptive gap is the primary motivation for the present thesis. The sequence features characterised by Faltejsková et al. (2026) extend over 1–2 kb on either side of the binding site — a spatial scale that is inaccessible to conventional motif-based models by design, but that falls within the context window of recent genomic foundation models operating at single-nucleotide resolution. Whether the per-token embeddings of such a model encode the kilobase-scale compositional structure described above is the empirical question taken up in this thesis. The foundation model used, Evo 2, is introduced in §1.4.

1.3 Computational approaches to TF binding site analysis

1.3.1 Position weight matrices

The dominant computational representation of TF binding specificity is the position weight matrix (PWM), *also called a position-specific scoring matrix (PSSM)*. A PWM is derived from a set of aligned binding sequences of fixed width L: at each position $i \in \{1,\dots,L\}$ and nucleotide $b \in \{A,C,G,T\}$, the log-odds score $w_{i,b} = \log_2(f_{i,b}/p_b)$ is computed, where $f_{i,b}$ is the observed frequency of nucleotide b at position i among the training sequences and p_b is the background frequency of b (Stormo, 2000). The score of a candidate sequence is the sum of per-position log-odds terms; a threshold on this score determines whether the sequence is called a binding site. The information content of the matrix, $\mathrm{IC} = \sum_i \sum_b f_{i,b} \log_2(f_{i,b}/p_b)$, is a measure of the specificity of the motif and is used to produce the familiar sequence logo visualisation (Schneider a Stephens, 1990). **[TODO: sequence logo visual]**

The PWM model rests on an independence assumption: the contribution of each position to the total score is treated as additive, with no coupling between positions. This assumption is biologically inaccurate — adjacent nucleotides are correlated in both sequence and structural terms — but the model is computationally tractable, transferable across organisms, and sufficiently accurate for many applications. It remains the standard representation in curated databases such as JASPAR (Ovek Baydar et al., 2026) and HOCOMOCO (Vorontsov et al., 2024), which together provide PWM models for the large majority of human TFs.

1.3.2 Motif discovery and enrichment analysis

Given a set of ChIP-seq peak sequences, two related tasks arise: discovering the sequence motif enriched in the peaks (motif discovery), and testing whether a known motif is significantly enriched relative to a background set (motif enrichment).

MEME (Multiple EM for Motif Elicitation) addresses the discovery task using expectation maximisation to identify the PWM that best explains an overrepresented pattern across the input sequences (Bailey a Elkan, 1995); the broader MEME Suite extends this to enrichment testing, scanning, and comparative analysis (Bailey, Johnson et al., 2015). HOMER (Hypergeometric Optimization of Motif EnRichment) is widely used for ChIP-seq specifically: it optimises motif enrichment relative to a GC-matched background set, making it robust to the GC bias that is characteristic of many TF binding sites (Heinz et al., 2010). Scanning a genome with a known PWM to identify all sites above a score threshold is performed by tools such as FIMO (Grant et al., 2011); this is the operation used to count the motif matches discussed in §1.1.3.

1.3.3 Extensions beyond the independence model

Several approaches have been developed to relax the positional independence assumption. Dinucleotide weight matrices (DWMs) model pairwise dependencies between adjacent positions, capturing the nearest-neighbour sequence correlations that are mechanistically important for shape readout (Siebert a Söding, 2016). Hidden Markov models and related probabilistic graphical models can represent variable-length motifs with internal positional dependencies, at the cost of increased model complexity and training data requirements. **[TODO: citation]** Convolutional neural networks trained on ChIP-seq data learn sequence representations that implicitly capture multi-position dependencies and outperform PWMs on held-out binding prediction benchmarks (Alipanahi et al., 2015); however, such models are typically trained in a supervised setting and require labelled positive and negative sequences.

A complementary line of work incorporates predicted DNA shape features alongside sequence features. Mathelier et al. (2016) showed that adding per-position shape predictions (minor groove width, propeller twist, roll, and helical twist computed by DNAshape) as features alongside the PWM score systematically improves in vivo binding prediction across a broad range of TF families. The deepDNAshape model (J. Li et al., 2024), which predicts the same structural features from sequence using a neural network trained on molecular dynamics simulations, is used by Faltejsková et al. (2026) to characterise shape changes across the 10 kb window around binding sites.

1.3.4 The kilobase-scale context gap

All methods described above share a fundamental constraint: they operate on a fixed window whose width is determined by the binding footprint of the TF. PWMs typically span 6–20 bp; even extended models incorporating shape features or positional dependencies are anchored to the core motif and its immediate neighbourhood, typically within 50 bp. By construction, these methods cannot represent the kilobase-scale sequence composition gradients described in §1.2.8, because they have no mechanism for incorporating sequence at that distance from the binding site.

This constraint is not a limitation of any specific algorithm but is inherent to the class of models: the binding site itself is defined as a short sequence, and the model is designed to score that sequence. Capturing kilobase-scale context requires a qualitatively different modelling approach — one that processes the full sequence window and produces representations that integrate information across its entire length. Genomic foundation models, described in the following section, are a natural candidate for this role.

1.4 Genomic foundation models

1.4.1 Self-supervised pretraining and the foundation model paradigm

The dominant paradigm in modern sequence modelling **[TODO: this is not specific to sequence modelling. bommasani is general about foundation model]** is self-supervised pretraining on large unlabelled corpora, followed by task-specific adaptation. A model is trained to predict held-out portions of the input sequence from the remaining context: in masked language modelling, randomly selected tokens are replaced with a mask symbol and the model is trained to recover them; in autoregressive modelling, the model is trained to predict each token from all preceding tokens. Neither objective requires labelled data; the supervision signal is derived entirely from the sequence itself. The resulting model learns distributed representations that encode statistical regularities of the training corpus and that can be transferred to downstream tasks by fine-tuning on small labelled datasets or by extracting intermediate-layer embeddings for use as features (Bommasani et al., 2022).

The transformer architecture (Vaswani et al., 2017) is the backbone of most current foundation models. Its core mechanism, scaled dot-product attention, computes pairwise interactions between all positions in the input sequence, allowing each position's representation to integrate information from arbitrarily distant context. This property makes transformers well-suited for tasks that require long-range dependencies; however, the computational and memory cost of full attention scales quadratically with sequence length, which places a practical ceiling on the context window at training time. For DNA, where biologically relevant interactions can span thousands to millions of base pairs, this is a fundamental constraint.

[TODO: Consider a schematic here contrasting the limited receptive field of a PWM scanner against the full-context attention of a transformer, to visually motivate why architecture matters for the kilobase-scale problem.]

1.4.2 Genomic foundation models: a brief landscape

The first genomic foundation model applied to regulatory sequence was DNABERT (Ji et al., 2021), which adapted the BERT masked language modelling objective to the human reference genome. DNABERT tokenises the genome into overlapping k-mers (default $k = 6$) and pretrains a bidirectional transformer on the resulting token sequences. The pretrained model was shown to transfer to promoter prediction, splice site detection, and TF binding site prediction by fine-tuning on small labelled datasets. A limitation of the k-mer tokenisation is that it conflates sequence position with token identity and produces a highly redundant vocabulary; the context window at inference time was also limited to 512 tokens, corresponding to roughly 512 bp of genomic sequence.

The Nucleotide Transformer (Dalla-Torre et al., 2025) scaled the masked language modelling approach to larger model sizes (up to 2.5 billion parameters) and a multi-species training corpus, demonstrating that scale improves transfer performance across a broad panel of regulatory prediction benchmarks. Its context window remains limited to a few kilobases at inference.

HyenaDNA (Nguyen, Poli, Faizi et al., 2023) addressed the context length problem by replacing the attention mechanism with the Hyena operator, a data-controlled convolutional recurrence that scales linearly in sequence length. Trained autoregressively on the human reference genome at single-nucleotide resolution, HyenaDNA extended the practical context window to up to 1 million base pairs, demonstrating for the first time that a genomic foundation model could be trained and evaluated at the scale of entire chromosomal regions. Performance on short-range benchmarks was competitive with transformer-based models; the primary advance was the demonstration that linear-scaling architectures could handle genomic context lengths that are infeasible for full attention.

1.4.3 Evo and Evo 2

Evo (Nguyen, Poli, Durrant et al., 2024) extended the long-context autoregressive paradigm to a multi-kingdom training corpus, pretraining a 7 billion parameter model on prokaryotic, viral, and eukaryotic genomes using the StripedHyena architecture. StripedHyena interleaves multi-head attention layers with Hyena convolutional layers, combining the strong local feature extraction of attention with the linear-scaling long-range context capture of the Hyena operator. Evo demonstrated capabilities in zero-shot variant effect prediction, coding sequence generation, and CRISPR guide RNA design.

Evo 2 (Brixi et al., 2026) is the successor model, published in *Nature* in 2026. It is trained autoregressively on OpenGenome2, a curated dataset of 8.8 trillion nucleotide tokens drawn from organisms across all domains of life, including prokaryotes, archaea, and eukaryotes. The architecture is StripedHyena 2, an updated version that further improves the integration of attention and gated convolutional layers. Models are available in three sizes: 1 billion, 7 billion, and 40 billion parameters. The 7 billion parameter checkpoint used in this thesis operates at single-nucleotide resolution with a context window of up to 1 million base pairs in its extended-context configuration (`evo2_7b`), or 8192 tokens in its base configuration (`evo2_7b_base`). Because the 10 kb windows used in this thesis exceed the 8192-token limit of the base checkpoint, the extended-context `evo2_7b` checkpoint is used throughout.

Evo 2 has been benchmarked on variant effect prediction across coding and non-coding variants, zero-shot fitness prediction for proteins and ncRNA, and generative sequence design; these evaluations are described in the original publication. Whether the per-token embeddings of Evo 2 encode the kilobase-scale flanking-sequence structure characterised by Faltejsková et al. has not been examined.

1.4.4 Embedding extraction and layer selection

Foundation models trained autoregressively produce, at each token position, a hidden-state vector whose dimensionality equals the model's embedding dimension. These per-token hidden states are the primary substrate for embedding-based analyses: they can be mean-pooled across the sequence to produce a single fixed-length representation, or retained at full resolution to carry positional information.

For Evo 2 with a 7 billion parameter model, the embedding dimension is 4096. Each token (nucleotide) at each sequence position produces a 4096-dimensional vector; a 10 kb sequence therefore yields a $(10{,}000 \times 4096)$ matrix of hidden states before any pooling.

The choice of which layer's hidden states to extract is consequential. The final layer is specialised for the next-token prediction objective and carries information optimised for that task rather than for general-purpose representation. Intermediate layers have been shown empirically to produce more transferable representations for downstream tasks; *the exon classifier notebook distributed with Evo 2 uses layer `blocks.28.mlp.l3` as the embedding source for a classification task* **[TODO: what else do they use it for?]**, and this layer is used in the present thesis for the same reason.

[TODO: consider cutting] In this thesis, each 10 kb sequence is processed in a single forward pass through `evo2_7b`, and the per-token hidden states from `blocks.28.mlp.l3` are extracted and mean-pooled within non-overlapping 50 bp bins to yield a (200×4096) positional embedding profile per peak. This binning resolution matches the 50 bp window size used by Faltejsková et al. (2026) for hand-engineered feature extraction, allowing a direct spatial correspondence between the two analyses. The technical details of the embedding pipeline are described in §**??**.

1.4.5 Autoregressive context and its implications

A property of autoregressive models that is directly relevant to the interpretation of positional embedding profiles is the asymmetry of context: at any token position t, the hidden state is computed from all preceding tokens $1, \ldots, t-1$ but has no access to tokens at positions $t+1, \ldots, L$. Tokens near the beginning of the sequence therefore have little or no left context, and their representations may differ systematically from tokens in the interior of the sequence where left context is rich. This warm-up effect is a model-level property, not a sequence-biological signal, and must be accounted for when interpreting positional embedding profiles derived from fixed-start 10 kb windows. The practical consequence for this thesis — and the trimming strategy adopted to address it — is described in §**??**.

[TODO: simplified StripedHyena 2 diagram, or the one from the paper]

Chapter 2

Aims of thesis

1.

Chapter 3

Methods

3.1 Data

3.1.1 Transcription factors

The descriptive analysis is performed on three human transcription factors selected to span distinct binding modes relevant to the questions posed in this thesis: CTCF, MYC, and SPI1. CTCF is an eleven-zinc-finger architectural factor with a long, well-characterised consensus motif and broad genomic occupancy, included as a strong-motif baseline against which embedding-based representations can be evaluated (Ong a Corces, 2014) MYC is a basic helix-loop-helix leucine-zipper factor that, in the analysis of Faltejsková et al. (2026), displays a pronounced directional dinucleotide asymmetry in the kilobase-scale flanks of its in vivo binding sites, making it a natural target for testing whether long-range sequence structure is reflected in Evo 2 representations (Lüscher a Vervoorts, 2012). SPI1 (PU.1) is an ETS-family lineage-specific factor that does not exhibit the directional flanking signature reported for MYC and is therefore included as a contrast factor for which the kilobase-scale signal is expected to be weaker or absent (Rothenberg et al., 2019).

3.1.2 ChIP-seq data sources

For each transcription factor, three ChIP-seq experiments are obtained from the ENCODE portal (Dunham et al., 2012), yielding nine BED files in total **[TODO: cite ENCODE portal / ENCODE3 if a more recent reference is preferred]**. Conservative IDR-thresholded narrowPeak files (Q. Li et al., 2011) are used where available; otherwise, the IDR-thresholded peak file is used. Cell-line selection follows the inclusion criterion adopted by Faltejsková et al. (2026), which retains only cell lines appearing in at least three experiments across their panel. The accessions, transcription factors, and biosamples comprising the embedded set are listed in Table **??**, alongside the number of peaks supplied to the embedding pipeline. Sequence is extracted from the human reference assembly hg19 to maintain consistency with Faltejsková et al. (2026) and to permit direct comparison of per-bin readouts in the sanity check described in Section **??**; this thesis does not consume any of the hg38-based annotation tracks

Tabulka 3.1: Chosen ChIP-seq experiments

TF	**Cell line**	**ENCODE accession**
MYC	GM12878	ENCFF765CKK
MYC	H1	ENCFF700CXD
MYC	MCF-7	ENCFF149BIS
CTCF	GM12878	ENCFF017XLW
CTCF	H1	ENCFF692RPA
CTCF	K562	ENCFF769AUF
SPI1	GM12878	ENCFF881ARS
SPI1	GM12878	ENCFF198SPL
SPI1	K562	ENCFF185DGL

(non-B DB, CpG islands, ENSEMBL regulatory features) used in the published study **[TODO: cite hg19 / GRCh37 (UCSC or Genome Reference Consortium)]**. BED files are downloaded programmatically from the ENCODE portal using the URL schema `https://www.encodeproject.org/files/{accession}/@@download/{accession}.bed.gz`. **[TODO: fix wrapping?]**

3.1.3 Cell-line coverage

The intended design covers the cell lines GM12878, H1, and K562 across all three transcription factors, providing a balanced grid for within-TF and across-TF comparisons. The realised design departs from this target in one respect: the MYC K562 file (ENCFF043WTJ) was not embedded within the available compute window, and the MYC MCF-7 file (ENCFF149BIS) was substituted as the third MYC experiment. The resulting cell-line coverage is therefore CTCF on GM12878, H1, K562; MYC on GM12878, H1, MCF-7; and SPI1 on GM12878 (two replicates) and K562. The interpretive consequences of this imbalance for any TF–cell-line interaction analysis are addressed in the Results and Discussion chapters.

3.1.4 Random-windows control

A random-windows control is constructed to provide a non-peak-centred reference that shares the window geometry, sequence preprocessing, and embedding pipeline of the ChIP-seq data and against which TF-centred positional structure can be contrasted. Two thousand 10 kb windows are sampled from hg19 autosomes and chromosome X, with chromosome selection weighted by usable sequence length and chromosomes Y and M excluded. Candidate windows are subjected to rejection sampling against the union of MYC and CTCF peak sets and against previously accepted control windows, ensuring the control is disjoint from those peak sets and internally non-overlapping. SPI1 peaks are not included in the rejection set: the control was generated before SPI1 was added to the design and was not subsequently regenerated; given the small genomic footprint of the SPI1 peak set the residual chance of overlap is bounded but non-zero. A fixed random seed of 42 is used throughout.

3.1.5 Reproducibility and seeding

Stochastic elements of the pipeline are seeded explicitly to support reproducibility of intermediate quantities. A fixed random seed of 42 is applied at every sampling and stochastic-fitting site, including peak subsampling, random-windows generation, and UMAP fitting; this differs from the procedure of Faltejsková et al. (2026), in which peak subsampling is performed without a fixed seed. The full source code, configuration files, and environment-setup scripts used to produce the embedded set and the analyses reported in subsequent chapters are available at the project repository `https://github.com/TypeTwo30312/tv-thesis`. **[TODO: probably footnote the repo]**

3.2 Pipeline

This part describes the deterministic chain from ENCODE BED files to per-peak embedding tensors stored on disk. The three stages—sequence extraction, embedding extraction, and the computational environment in which they run—are each implemented as a separate Jupyter notebook in the project repository, and are reported here in the order in which they execute.

3.2.1 Sequence extraction

Window extraction reads an ENCODE narrowPeak file and outputs a per-peak record containing the peak coordinates and a fixed-length DNA sequence centred on the binding site. The 10 kb window length is chosen to match the ±5000 bp range over which Faltejsková et al. (2026) report statistically significant compositional structure around in vivo binding sites. The peak midpoint is used as the binding-site proxy: for the ENCODE narrowPeak files inspected in this thesis, the reported summit coincides with the midpoint of the peak interval (CTCF GM12878 peaks are uniformly ~356 bp wide, MYC GM12878 peaks ~390 bp, MYC H1 ~264 bp), and the midpoint convention matches that used by Faltejsková et al. (2026). Peaks whose 10 kb window would extend past a chromosome boundary are dropped rather than padded, since padding with non-biological characters would introduce a fixed-position bias into any subsequent positional analysis. Non-ACGT characters in the extracted sequence are masked to N at the extraction stage; the analysis-stage filtering of windows containing fully-N 50 bp bins is described in Part 3 (Section **??**). **[TODO: adjust ?? when Part 3 is drafted]** For each BED file, peaks are uniformly subsampled to a maximum of 10 000 (all peaks are retained if fewer are available), matching the `MAX_SIZE_SAMPLED` parameter used by Faltejsková et al. (2026) but with a fixed random seed of 42 throughout. Each output row carries the columns `peak_id`, `chr`, `start`, `end`, `center`, and `sequence`, and is written as a gzip-compressed CSV. The `ACCESSION__TF__BIOSAMPLE` naming convention used throughout the pipeline is preserved from the published source code released alongside Faltejsková et al. (2026), so that the per-bin feature-extraction scripts from that work remain directly applicable to the sanity check reported in Part 3 (Section **??**). **[TODO: adjust ?? when Part 3 is drafted]**

3.2.2 Embedding extraction

Embedding extraction is performed with the Evo 2 7B checkpoint released by Brixi et al. (2026); the architecture and pretraining objective are described in the background chapter (Section **??**). **[TODO: adjust ?? to the actual background-chapter label]** The 7B family is used because it runs natively in bfloat16 on Ampere-class hardware and does not require Transformer Engine or FP8 support, which are needed for the 20B and 40B variants and are unavailable on the GPUs accessible through the CERIT-SC cluster used in this work. **[TODO: not true!! fix, reasoning]** Within the 7B family, the 1 M-context variant (`evo2_7b`) is selected over the 8 kb-context base variant (`evo2_7b_base`) because the chosen 10 kb window exceeds the latter's context length. A single intermediate-layer embedding is extracted per token from the layer `blocks.28.mlp.l3`; intermediate layers are recommended over the final layer for representation learning because the final layer is specialised to the next-token prediction objective, and `blocks.28.mlp.l3` is the layer used by the exon-classifier example in the official Evo 2 repository (Brixi et al., 2026). **[TODO: either run a comparison or suggest it in discussion]** Each 10 kb sequence is tokenised under the Evo 2 byte-level scheme into a $(1, 10000)$ integer tensor (one token per nucleotide), passed through the model in a single forward pass on one GPU, and the resulting $(1, 10000, 4096)$ bfloat16 activation tensor is reshaped into 200 non-overlapping 50 bp bins and averaged within each bin to yield a $(200, 4096)$ representation per peak, which is cast to fp16 before writing. The 50 bp bin width matches the k-mer resolution adopted by Faltejsková et al. (2026) so that per-bin readouts are directly comparable; non-overlapping bins are used in place of their 25 bp stride because each bin already integrates 50 input positions of the same forward pass and stride overlap would not contribute additional information. Unlike the strictly local hand-engineered features used by Faltejsková et al. (2026), each binned embedding has the full 10 kb window in its causal receptive field via the autoregressive forward pass, and so reflects model-internal aggregation of context up to that point rather than a fixed-window feature of the sequence itself. A streaming-write strategy based on `numpy.memmap` is used for the per-file accumulator together with a checkpoint every 500 peaks, because retaining all per-peak tensors in memory before writing exceeds the per-pod RAM allocation on the cluster for the larger BED files. The output of this stage is, per BED file, a single `.npz` archive containing the $(n_{\text{peaks}}, 200, 4096)$ fp16 array together with a `.parquet` sidecar carrying the `peak_id` column and BED-derived metadata in the same row order. Embedding extraction was performed on a single NVIDIA A40 GPU (46 GB VRAM) with peak allocation of 19.17 GB at batch size 1, at an observed throughput of approximately 0.5 peaks per second; the embedded set comprises the 9 ChIP-seq files of Part 1 plus the random-windows control of Section 3.1.4, totalling 96.9 GB of compressed embeddings on disk.

3.2.3 Computational environment

The embedding stage runs on the CERIT-SC JupyterHub platform under the NVIDIA PyTorch 2.5.0 container image, which provides a Python 3.10, CUDA 12.6, and Ubuntu 22.04 base; a project-specific Python 3.12 virtual environment is layered on top, installed via uv into the persistent home directory and pinning `torch` 2.7.1+cu128, `flash-attn` 2.8.0.pos and `evo2`. A separate CPU-only environment is used for the analyses reported in Part 3, isolating the heavy GPU stack from sessions that do not require it and pinning `numpy`, `polars`, `scikit-learn`, and `umap-learn` alongside the standard scientific Python packages. Both environments are reproduced by setup scripts (`setup_evo2_env.sh`, `setup_exploratio` **[TODO: nope! single evo2 environment, no setup scripts]** maintained alongside the source code, and the full code base, configuration files, and notebooks are available at the project repository[1].

3.3 Analyses

[1] `https://github.com/TypeTwo30312/tv-thesis`

Chapter 4

Results and discussion

4.1 Podkapitola

Závěr

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Appendix

Sem můžete přidat přílohu. Pokud chcete "Přílohy", tak upravte definici záhlaví v souboru sci.muni.thesis.sty, viz příkaz `\HlavickaPriloha`**.**

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